Volumes of Revolution

ANSWER KEY

Section 1: Disk Method (Rotation Around Axes)

1 Find the volume of the solid generated by revolving $y = \sqrt{x}$ from $x = 0$ to $x = 4$ about the x-axis.

a. 8π ✓

b. 16π

c. 4π

d. $32\pi/3$

Disk method: $V = \pi\int_0^4 [f(x)]^2\,dx = \pi\int_0^4 x\,dx = \pi[x^2/2]_0^4 = \pi(16/2) = 8\pi$.

2 Find the volume when the region under $y = x^2$ from $x = 0$ to $x = 2$ is rotated about the x-axis.

a. $\frac{32\pi}{5}$ ✓

b. $\frac{16\pi}{5}$

c. 8π

d. $\frac{64\pi}{5}$

$V = \pi\int_0^2 (x^2)^2\,dx = \pi\int_0^2 x^4\,dx = \pi[x^5/5]_0^2 = \frac{32\pi}{5}$.

3 Find the volume when the region between $x = \sqrt{y}$, $y = 0$, and $y = 4$ is revolved about the y-axis.

a. 8π ✓

b. 16π

c. 4π

d. $32\pi/3$

Disk method with respect to y: $V = \pi\int_0^4 [\sqrt{y}]^2\,dy = \pi\int_0^4 y\,dy = \pi[y^2/2]_0^4 = 8\pi$.

Section 2: Washer Method (Hollow Solids)

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4 Find the volume when the region bounded by $y = x$ and $y = x^2$ is revolved about the x-axis.

a. $\frac{2\pi}{15}$ ✓

b. $\frac{\pi}{15}$

c. $\frac{4\pi}{15}$

d. $\frac{\pi}{5}$

Outer $R = x$, inner $r = x^2$. $V = \pi\int_0^1 (x^2 - x^4)\,dx = \pi[x^3/3 - x^5/5]_0^1 = \pi(1/3 - 1/5) = \frac{2\pi}{15}$.

5 The region bounded by $y = 4 - x^2$ and $y = 0$ is rotated about the line $y = -1$. Set up the outer and inner radii.

a. $R(x) = (4 - x^2) - (-1) = 5 - x^2$, $r(x) = 0 - (-1) = 1$ ✓

b. $R(x) = 4 - x^2$, $r(x) = 1$

c. $R(x) = 5 - x^2$, $r(x) = 0$

d. $R(x) = 4 - x^2 + 1$, $r(x) = 2$

Distance from curve to axis of revolution $y = -1$: outer radius $R(x) = (4 - x^2) - (-1) = 5 - x^2$, inner radius $r(x) = 0 - (-1) = 1$.

6 Evaluate the volume in the previous problem rotated about $y = -1$ on $[-2, 2]$.

a. $\frac{384\pi}{15} = \frac{128\pi}{5}$ ✓

b. $\frac{64\pi}{5}$

c. 32π

d. $\frac{256\pi}{5}$

$V = \pi\int_{-2}^{2}[(5 - x^2)^2 - 1^2]\,dx = 2\pi\int_0^2 (24 - 10x^2 + x^4)\,dx = 2\pi[24x - \frac{10}{3}x^3 + \frac{x^5}{5}]_0^2 = 2\pi(48 - 80/3 + 32/5) = 2\pi(\frac{7}{2}$
$24(2) = 48$; $2\pi(48 - 80/3 + 32/5) = 2\pi(\frac{416}{15}) = \frac{832\pi}{15}$.

Section 3: Cylindrical Shells Method

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7 State the cylindrical shells formula for a region under $y = f(x)$ on $[a, b]$ revolved about the y-axis.

a. $V = 2\pi\int_a^b xf(x)\,dx$ ✓

b. $V = \pi\int_a^b [f(x)]^2\,dx$

c. $V = 2\pi\int_a^b [f(x)]^2\,dx$

d. $V = \pi\int_a^b x^2 f(x)\,dx$

Circumference $= 2\pi x$, height $= f(x)$, thickness $= dx$. Shell volume $= 2\pi\int_a^b xf(x)\,dx$.

8 Find the volume when the region under $y = 3x - x^2$ from $x = 0$ to $x = 3$ is revolved about the y-axis.

a. $\frac{27\pi}{2}$ ✓

b. 27π

c. $\frac{81\pi}{4}$

d. $\frac{9\pi}{2}$

Shells:
$V = 2\pi\int_0^3 x(3x - x^2)\,dx = 2\pi\int_0^3 (3x^2 - x^3)\,dx = 2\pi[x^3 - \frac{x^4}{4}]_0^3 = 2\pi(27 - 81/4) = 2\pi(27/4) = \frac{27\pi}{2}$.

9 Use cylindrical shells to find the volume when $y = e^{-x^2}$ on $[0, 1]$ is revolved about the y-axis.

a. $\pi(1 - e^{-1})$ ✓

b. $2\pi(1 - e^{-1})$

c. πe^{-1}

d. $\frac{\pi}{2}(1 - e^{-1})$

$V = 2\pi\int_0^1 xe^{-x^2}\,dx$. Let $u = -x^2$, $du = -2x\,dx$. $V = -\pi\int_0^{-1} e^u\,du = \pi\int_{-1}^0 e^u\,du = \pi[e^u]_{-1}^0 = \pi(1 - e^{-1})$.

Section 4: Cross-Sections with Known Geometric Shapes

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10 The base of a solid is the circle $x^2 + y^2 = 4$. Cross-sections perpendicular to the x-axis are squares. Find the volume.

a. $\frac{128}{3}$ ✓

b. 16π

c. 32

d. $\frac{64}{3}$

Side of square $s = 2y = 2\sqrt{4 - x^2}$. Area $A(x) = s^2 = 4(4 - x^2)$.
$V = \int_{-2}^{2} 4(4 - x^2)\,dx = 8\int_{0}^{2} (4 - x^2)\,dx = 8[4x - x^3/3]_0^2 = 8(8 - 8/3) = 8(16/3) = 128/3$.

11 The base of a solid is bounded by $y = x^2$ and $y = 4$. Cross-sections perpendicular to the y-axis are semicircles. Find the volume.

a. 4π ✓

b. 2π

c. 8π

d. $\frac{8\pi}{3}$

Diameter $d = 2x = 2\sqrt{y} \Rightarrow r = \sqrt{y}$. Area $A(y) = \frac{1}{2}\pi r^2 = \frac{1}{2}\pi y$. $V = \int_0^4 \frac{1}{2}\pi y\,dy = \frac{\pi}{2}[y^2/2]_0^4 = \frac{\pi}{2}(8) = 4\pi$.

12 The base is the triangle with vertices $(0, 0), (1, 0), (0, 1)$. Cross-sections perpendicular to the x-axis are equilateral triangles. Find the volume.

a. $\frac{\sqrt{3}}{12}$ ✓

b. $\frac{\sqrt{3}}{6}$

c. $\frac{1}{6}$

d. $\frac{\sqrt{3}}{4}$

Side $s = 1 - x$. Area of equilateral triangle $A(x) = \frac{\sqrt{3}}{4}s^2 = \frac{\sqrt{3}}{4}(1 - x)^2$.
$V = \frac{\sqrt{3}}{4}\int_0^1 (1 - x)^2\,dx = \frac{\sqrt{3}}{4}[-\frac{(1-x)^3}{3}]_0^1 = \frac{\sqrt{3}}{4}(1/3) = \frac{\sqrt{3}}{12}$.

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